VAC – AI SKILLS COURSE

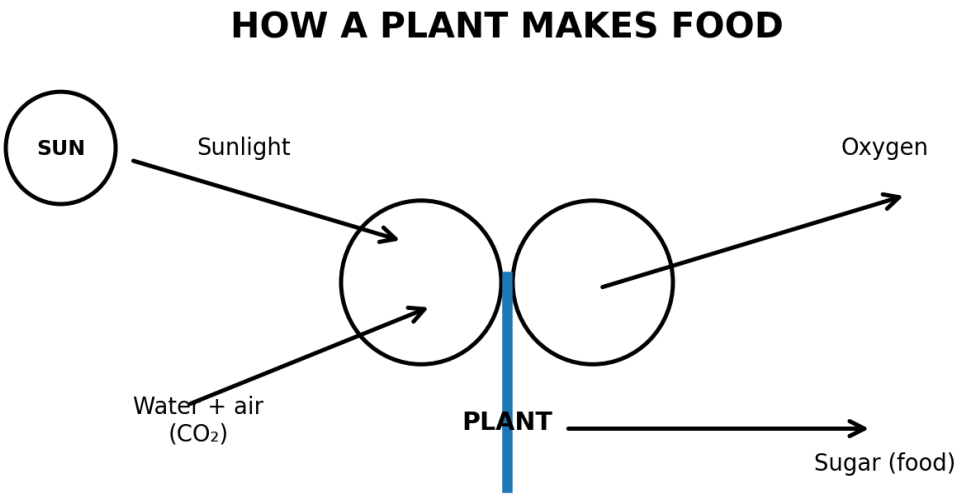
CAPSTONE 5

AI CONCEPT EXPLAINER

One Topic at Three Levels

Topic: PHOTOSYNTHESIS

A study of how AI can explain one scientific concept to a school student, an undergraduate learner and an expert without losing scientific accuracy.



Plants use light, water and carbon dioxide to make sugar and release oxygen.

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Academic Project Report | 2026

1. Introduction and Project Objective

Photosynthesis is the process by which plants, algae and some bacteria capture light energy and use it to build energy-rich organic molecules. In green plants, it mainly takes place in chloroplasts. This project uses AI to explain the same concept at three levels while keeping the central scientific facts consistent.

Project question: How well can AI explain a difficult concept for a child, a beginner and an expert, without losing accuracy?

Objectives

- Prepare three explanations for different learning levels.
- Use analogies and a visual for each level.
- Check the explanations against reliable educational sources.
- Prepare a short explainer-video script and quiz.
- Compare an initial human draft with the AI-assisted version.

2. Concept Background

The overall simplified equation is: **6CO■ + 6H■O + light energy** → **C■H■■O■ + 6O■**. The equation summarizes the inputs and outputs, but the real process happens through many enzyme-controlled steps. In plants, the light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH. The Calvin cycle takes place in the stroma and uses ATP, NADPH and carbon dioxide to make G3P, a molecule used to build carbohydrates.

Important terms: chloroplast – the cell organelle where photosynthesis occurs; chlorophyll – a light-absorbing pigment; ATP and NADPH – temporary energy/electron carriers; Calvin cycle – the carbon-fixation pathway.

3. Explanation for Class 8 Student

Prompt used: “Explain photosynthesis to a Class 8 student in simple English. Use short sentences, avoid difficult terms, include the inputs and outputs, and do not introduce incorrect science.”

Explanation:

Plants make their own food through a process called photosynthesis. Their leaves take in carbon dioxide from the air, and their roots absorb water from the soil. A green substance in the leaves called chlorophyll catches energy from sunlight. The plant uses this energy to turn water and carbon dioxide into sugar. The sugar helps the plant grow and live. Oxygen is also released into the air. So, sunlight, water and carbon dioxide are used to make sugar and oxygen.

Analogy: Think of a leaf as a tiny solar-powered kitchen. Sunlight provides the energy, water and carbon dioxide are the ingredients, and the plant makes sugar as food. The analogy is only a learning aid: a leaf is not literally a kitchen, and photosynthesis is a series of chemical reactions.

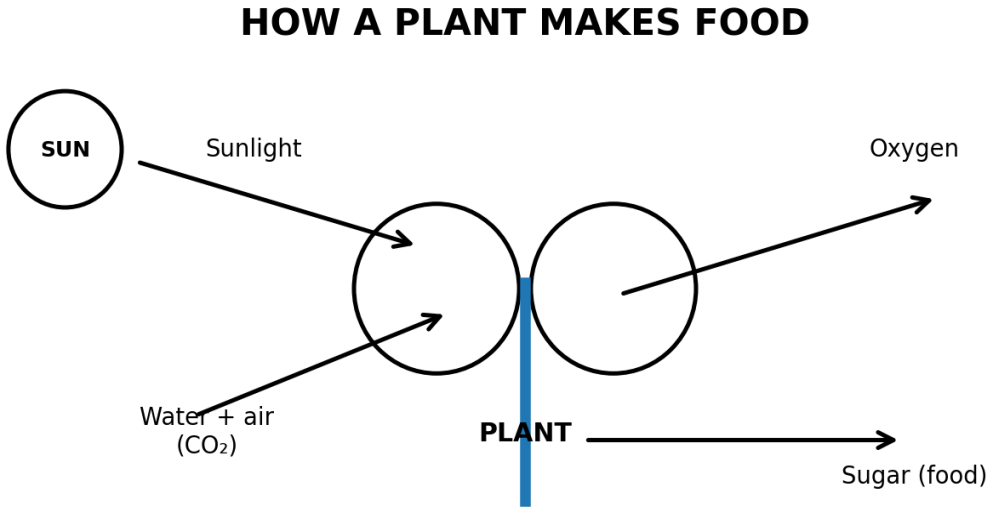


Figure 1. Class 8 visual: sunlight, water and carbon dioxide enter the process; sugar and oxygen are produced.

4. Explanation for an Undergraduate Beginner

Prompt used: “Explain photosynthesis at introductory undergraduate level. Include the chloroplast, thylakoid membrane, stroma, light-dependent reactions, ATP, NADPH and Calvin cycle. Explain the relationship between the stages accurately.”

Explanation:
Photosynthesis converts light energy into chemical energy stored in organic molecules. In plants, it occurs mainly in chloroplasts. Chlorophyll and other pigments absorb light in the thylakoid membranes. During the light-dependent reactions, water is used, oxygen is released, and ATP and NADPH are produced. These energy-carrying molecules are then used in the Calvin cycle, which occurs in the stroma. The Calvin cycle fixes carbon dioxide and produces G3P, a three-carbon carbohydrate precursor. G3P can be used to build glucose and other carbohydrates. The two stages are linked because the Calvin cycle depends on ATP and NADPH made by the light-dependent reactions.

Analogy: The light reactions are like charging reusable energy packs (ATP and NADPH). The Calvin cycle uses those charged packs to assemble carbon-based building blocks. The analogy does not mean ATP and NADPH are batteries; they are molecules that transfer energy and reducing power.

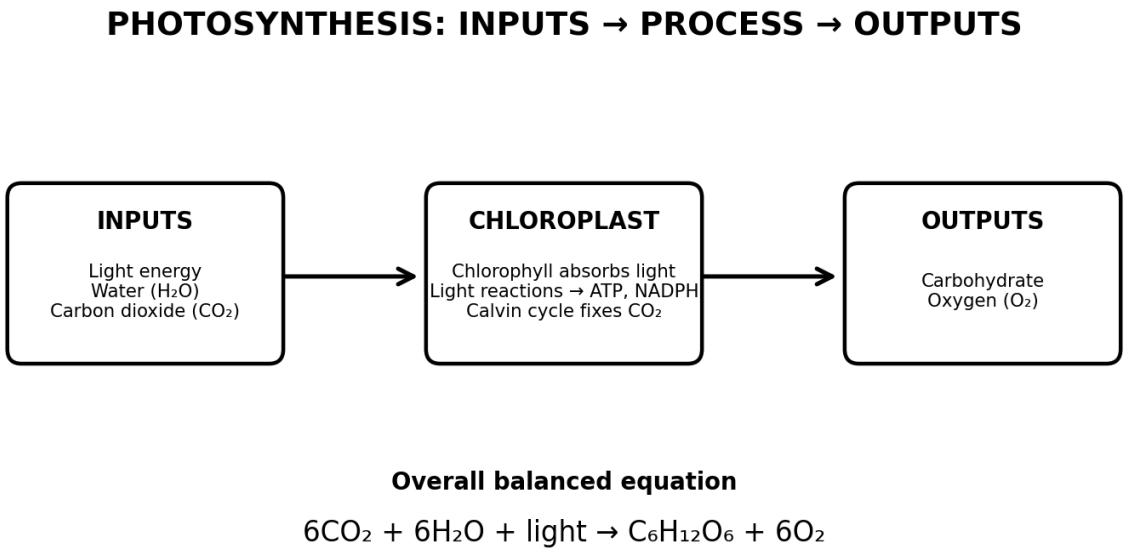


Figure 2. Undergraduate visual: inputs pass through chloroplast processes to produce carbohydrates and oxygen.

5. Explanation for an Expert / Advanced Learner

Prompt used: “Give an advanced, concise explanation of oxygenic photosynthesis in plants. Distinguish the light-dependent electron-transport reactions from the Calvin–Benson cycle; specify locations, key energy carriers and carbon-fixation product. Avoid implying that the Calvin cycle occurs only at night.”

Explanation:
In oxygenic photosynthesis, photosynthetic electron transport in the thylakoid membrane converts absorbed photon energy into chemical potential. Photosystem II extracts electrons from water, releasing molecular oxygen and contributing to a proton gradient. Electron transfer through the photosynthetic chain supports proton-motive-force-driven ATP synthesis, while photosystem I enables reduction of NADP■ to NADPH. In the chloroplast stroma, the Calvin–Benson cycle uses ATP and NADPH to assimilate CO■. Rubisco catalyses the carboxylation of ribulose-1,5-bisphosphate, producing 3-phosphoglycerate; subsequent reactions yield triose phosphate, including G3P, while regenerating the CO■ acceptor. Triose phosphate can contribute to carbohydrate synthesis.

Analogy: The thylakoid system is a coupled energy-conversion line: photon-driven electron transfer establishes conditions for ATP synthesis and NADPH formation. The Calvin–Benson cycle is a carbon-assimilation network that spends this chemical energy to reduce and incorporate inorganic carbon. This analogy describes functional coupling, not a literal factory.

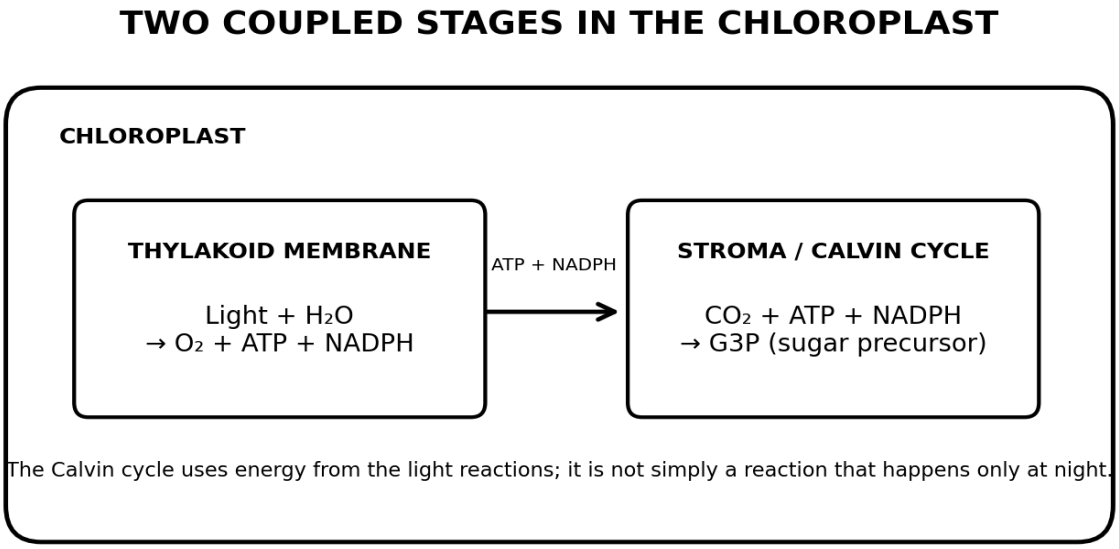


Figure 3. Advanced visual: the two linked stages and their locations within the chloroplast.

6. Verification Notes and Oversimplifications

The three versions were checked against OpenStax Biology 2e (Sections 8.1–8.3) and Monash University's biology learning resource. The core facts below remain consistent across all levels.

Claim checked	Verification / correction
Inputs and outputs	Light energy, CO■ and H■O support carbohydrate production; O■ is released. The balanced equation is a summary, not a single-step reaction.
Location	In plants, photosynthesis occurs in chloroplasts; light reactions occur in thylakoid membranes and the Calvin cycle in the stroma.
Two stages	Light reactions produce ATP and NADPH; the Calvin cycle uses them to fix CO■ and produce G3P.
Oxygen source	The released O■ comes from water splitting in the light-dependent reactions, not from CO■.
"Dark reaction" wording	Calling the Calvin cycle a "dark reaction" can mislead. It does not directly use light as a reactant, but it depends on products of light reactions and is not simply a night-only process.

Oversimplifications identified and managed

1. "Plants eat sunlight" is not scientifically precise. Better: plants capture light energy and store it as chemical energy.
2. "Plants make glucose directly in one step" hides the multi-step pathway. At advanced level, the Calvin cycle's direct carbohydrate product is triose phosphate such as G3P.
3. "The Calvin cycle happens in darkness" is misleading. It is light-independent in its direct requirements, but it relies on products of light reactions and is regulated by light.
4. The equation may suggest oxygen comes from carbon dioxide. In oxygenic photosynthesis, released oxygen originates from water.

7. AI Tools and Skills Used

Tools selected (free tiers where available):

• ChatGPT – draft explanations, prompt refinement and quiz creation.
• Gemini – generate an independent explanation for comparison.
• Perplexity – locate and cross-check educational references.
• Canva – arrange the three diagrams into simple presentation slides.
• CapCut or ElevenLabs – record/edit narration and add voice-over (check current free-tier limits).

AI skills demonstrated: AI foundations, prompt engineering, content writing, research and verification, image/visual design, presentation design, and audio/video production.

Note: This report provides a practical workflow and prepared content. The student should actually run the prompts in the selected tools, record the voice-over/video, and retain screenshots or evidence of tool use if the instructor asks for proof.

8. Explainer Video Plan (2–3 Minutes)

Suggested title: Photosynthesis Explained at Three Levels

Slide 1 – Introduction (0:00–0:20)
“Have you ever wondered how a plant gets its food? Plants make much of their own food using a process called photosynthesis. Let us understand it at three different levels.”

Slide 2 – Class 8 level (0:20–0:55)
“A plant uses sunlight, water and carbon dioxide to make sugar. Chlorophyll in the leaves absorbs light. Oxygen is released into the air. Imagine a leaf as a solar-powered kitchen: sunlight supplies energy and the plant makes sugar.”

Slide 3 – Undergraduate level (0:55–1:40)
“Photosynthesis takes place in chloroplasts. The light-dependent reactions happen in the thylakoid membranes and produce ATP and NADPH. The Calvin cycle happens in the stroma and uses these molecules to fix carbon dioxide and produce G3P, which can be used to make carbohydrates.”

Slide 4 – Expert level (1:40–2:25)
“In oxygenic photosynthesis, photosystem II extracts electrons from water and releases oxygen. Electron transport supports ATP synthesis, and photosystem I contributes to NADPH formation. The Calvin–Benson cycle uses ATP and NADPH to assimilate carbon dioxide through Rubisco-mediated carbon fixation.”

Slide 5 – Conclusion (2:25–2:45)
“The language becomes more technical as the learner’s level increases, but the central facts remain the same: light energy is converted into chemical energy, carbon dioxide is fixed into organic molecules, and oxygen is released.”

Production steps: Create five slides in Canva, add the three diagrams, record or generate a permitted voice-over, import slides and audio into CapCut, add captions, check pronunciation of CO■, ATP, NADPH and Rubisco, then export a 2–3 minute video.

9. Understanding Quiz for Three Volunteers

Ask three classmates/volunteers to answer the same short quiz after watching the explainer. Use the table to record real results; do not fill in invented scores.

No.	Question	Expected answer
1	What three things are needed for photosynthesis?	Light, water and carbon dioxide.
2	Which pigment absorbs light in green plants?	Chlorophyll.
3	Where do light-dependent reactions occur?	Thylakoid membranes.
4	What do ATP and NADPH do?	They transfer energy/reducing power used in carbon fixation.
5	What is the direct carbohydrate product of the Calvin cycle?	G3P / triose phosphate.

Volunteer	Score / 5	What was easy?	What needs clarification?
Volunteer 1	Pending - actual volunteer quiz results		
Volunteer 2	Pending - actual volunteer quiz results		

Volunteer 3	Pending - actual volunteer quiz results		

10. Comparison: First Draft vs AI-Assisted Version

My first draft (sample baseline): “Plants take sunlight and make food. They take in water and air and give out oxygen.”

AI-assisted improved version: “In green plants, chlorophyll absorbs light energy in chloroplasts. The plant uses water and carbon dioxide to build carbohydrates through light-dependent reactions and the Calvin cycle. Oxygen is released during the light-dependent reactions when water is split.”

Comparison:

• Accuracy: The first draft gives a basic idea but leaves out the role of chlorophyll and the chloroplast. The improved version identifies the main structures and stages.

• Completeness: The first draft does not identify carbon dioxide clearly or explain how sugar is produced. The improved version names the inputs and the linked stages.

• Audience fit: The improved expert wording is more accurate for advanced learners, but it would be too technical for a Class 8 student. The three-level approach solves this by changing vocabulary and detail, not the underlying facts.

• Limitation: AI-generated content can sound confident while being inaccurate. Textbook or official-source checking is still necessary.

11. Prompt Library

1. Class 8: “Explain photosynthesis in 100–120 simple words for a Class 8 student. Include sunlight, water, carbon dioxide, chlorophyll, sugar and oxygen. Use one analogy and avoid misleading statements.”

2. Undergraduate: “Explain photosynthesis for a first-year undergraduate. Include chloroplast structure, thylakoid membrane, stroma, light-dependent reactions, ATP, NADPH and the Calvin cycle. Explain how the stages depend on one another.”

3. Expert: “Explain oxygenic photosynthesis at advanced level. Include PSII, water oxidation, electron transport, proton gradient, ATP synthase, PSI, NADP■ reduction, Rubisco and G3P. Distinguish established facts from simplifications.”

4. Verification: “Check the following explanation against OpenStax Biology 2e. List each factual claim, whether it is supported, any misleading simplification, and a corrected sentence. Do not invent references.”

5. Quiz: “Create five short questions that test understanding of photosynthesis from basic to advanced level. Provide an answer key separately.”

12. Conclusion

This project shows that AI can explain one concept in different ways for different audiences. A school-level explanation uses familiar words and an everyday analogy. An undergraduate explanation introduces organelles and the two main stages. An expert explanation includes electron transport, energy carriers and carbon fixation. Accuracy is maintained by checking key claims against trusted biology sources and by clearly identifying where analogies simplify the real process. AI is useful as a learning and drafting assistant, but human verification remains essential.

13. References

1. OpenStax. *Biology 2e*, Section 8.1: Overview of Photosynthesis.
https://openstax.org/books/biology-2e/pages/8-1-overview-of-photosynthesis

2. OpenStax. *Biology 2e*, Section 8.2: The Light-Dependent Reactions of Photosynthesis.
https://openstax.org/books/biology-2e/pages/8-2-the-light-dependent-reactions-of-photosynthesis

3. OpenStax. *Biology 2e*, Section 8.3: Using Light Energy to Make Organic Molecules.
https://openstax.org/books/biology-2e/pages/8-3-using-light-energy-to-make-organic-molecules

4. Monash University. *Photosynthesis – Student Academic Success*. https://www.monash.edu/student-academic-success/biology/photosynthesis

5. U.S. Department of Energy. *The Harnessed Atom – Teachers' Edition*. Photosynthesis. Educational reference PDF

Submission checklist: Add your name/register number; run and save evidence of prompts in at least four AI tools; record the 2–3 minute video; conduct the quiz with three real volunteers and enter their scores; attach the final video or its QR/link if required by your instructor.